

PROJECT PLANNING PHASE

Date: 19 June 2026

Team ID: -----

Project Name: AI-Enhanced Intrusion Detection System

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Dataset Collection	USN-1	As a developer, I can collect the NSL-KDD dataset for intrusion detection training.	3	High	Sourabh Patil
Sprint-1	Data Preprocessing	USN-2	As a developer, I can clean and preprocess network traffic data for machine learning.	5	High	Sourabh Patil
Sprint-1	Feature Engineering	USN-3	As a developer, I can select relevant features for intrusion detection.	3	High	Sourabh Patil
Sprint-2	Model Development	USN-4	As a developer, I can implement the Random Forest Classifier.	5	High	Sourabh Patil
Sprint-2	Model Training	USN-5	As a developer, I can train the model using NSL-KDD data.	5	High	Sourabh Patil
Sprint-2	Model Evaluation	USN-6	As a developer, I can evaluate the model using	4	High	Sourabh Patil

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			accuracy, precision, recall, and F1-score.			
Sprint-3	Flask Backend	USN-7	As a user, I can submit network traffic data for intrusion prediction.	5	High	Sourabh Patil
Sprint-3	Prediction Module	USN-8	As a user, I can receive attack classification results with confidence scores.	5	High	Sourabh Patil
Sprint-3	REST API	USN-9	As a developer, I can access prediction services through REST APIs.	4	Medium	Sourabh Patil
Sprint-4	Dashboard	USN-10	As a user, I can view intrusion statistics and monitoring charts.	5	High	Sourabh Patil
Sprint-4	Alert Logging	USN-11	As a user, I can view logged intrusion alerts and attack records.	4	High	Sourabh Patil
Sprint-4	Deployment & Testing	USN-12	As a developer, I can deploy and test the complete intrusion detection system.	6	High	Sourabh Patil

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	11	7 Days	01 June 2026	05 June 2026	11	05 June 2026
Sprint-2	14	7 Days	05 June 2026	10 June 2026	14	10 June 2026
Sprint-3	14	7 Days	11 June 2026	15 June 2026	14	15 June 2026
Sprint-4	15	7 Days	15 June 2026	19 June 2026	15	19 June 2026

Velocity Calculation

Total Story Points = 54

Sprint Duration = 18 Days

Average Velocity (AV)

AV = Total Story Points / Total Sprint Duration

AV = 54 / 28

AV = 1.93 Story Points per Day

Average Team Velocity ≈ 2 Story Points per Day

Burndown Chart Data

Day	Remaining Story Points
Day 1	54
Day 5	45
Day 10	36
Day 15	26
Day 19	17

The burndown chart shows a gradual reduction in remaining work throughout the project lifecycle until all 54 story points are completed by the final sprint.

Conclusion

The project planning phase establishes a structured Agile development process for the AI-Enhanced Intrusion Detection System. The project is divided into four sprints covering dataset preparation, machine learning model development, web application implementation, dashboard creation, alert management, testing, and deployment. The estimated velocity and sprint schedule ensure timely delivery while maintaining quality and scalability.